■ Great Barrier Reef Health Monitor

Complete Tech Stack Reference • SDG 14: Life Below Water

Next.js + FastAPI + CrewAI + RAG + Llama • Total Monthly Cost: $0

Architecture Overview

The system is split into two independently deployed services: a **Next.js frontend** hosted on Vercel and a **FastAPI backend** running inside a Docker container on HuggingFace Spaces. The backend orchestrates CrewAI agents powered by Llama 3.1 70B via Groq's free API, and uses a local RAG pipeline (sentence-transformers + ChromaDB) to query indexed reef reports. All data sources — NOAA, AIMS, GBRMPA, BOM, NASA — are free and open.

Layer	Technology	Hosted On	Cost
Frontend	Next.js 14 + TypeScript	Vercel	FREE
Backend	FastAPI + Uvicorn	HuggingFace Spaces (Docker)	FREE
LLM (primary)	Llama 3.1 70B via Groq	Groq Cloud	FREE tier
LLM (local dev)	Llama 3.1 8B via Ollama	Your machine	FREE
Embeddings	sentence-transformers MiniLM	Local CPU in container	FREE
Vector DB	ChromaDB (persistent)	HF Spaces 50 GB disk	FREE
Agents	CrewAI	Inside FastAPI container	FREE
Data Sources	NOAA / AIMS / GBRMPA / BOM / NASA	Public APIs	FREE

1. Frontend Stack (Deployed → Vercel)

1a. Core Framework

Technology	Version	Purpose	Cost
Next.js	14 (App Router)	React framework, SSR, routing	FREE
React	18	UI component library	FREE
TypeScript	5.x	Type safety — professional code quality	FREE
Node.js	20 LTS	Runtime	FREE

1b. Styling & UI Components

Technology	Purpose	Cost
Tailwind CSS	Utility-first CSS — fast, consistent styling	FREE
shadcn/ui	Pre-built professional components (cards, badges, alerts)	FREE
Lucide React	Icon library	FREE
Framer Motion	Smooth animations and page transitions	FREE

1c. Data Visualisation

Technology	Purpose	Cost
Recharts	Time-series charts: temperature trends, bleaching history	FREE
React Leaflet	Interactive reef map with zone-level health overlays	FREE
Leaflet.js	Underlying mapping library	FREE

Key UI element: React Leaflet renders a clickable map of the Great Barrier Reef with colour-coded bleaching alert zones — the most visually impressive demo piece.

1d. State Management & Data Fetching

Technology	Purpose	Cost
TanStack Query (React Query)	Server state management, API response caching	FREE
Zustand	Lightweight global state (active alerts, selected reef zone)	FREE
Axios	HTTP client for backend REST API calls	FREE
Socket.io-client	WebSocket connection — live agent progress updates	FREE

1e. Frontend Deployment

Platform	Details	Cost
Vercel	1-click deploy from GitHub, auto CI/CD, custom domain, 100 GB bandwidth, never sleeps	FREE
Vercel Analytics	Page views, performance metrics	FREE

2. Backend Stack (Deployed → HuggingFace Spaces Docker)

HuggingFace Spaces (Docker) specs: 16 GB RAM • 2 vCPUs • 50 GB persistent disk • Never sleeps • Free forever • Port 7860

2a. Web Framework

Technology	Version	Purpose	Cost
FastAPI	0.111+	REST API + WebSocket server, async support	FREE
Uvicorn	Latest	ASGI server — runs FastAPI	FREE
Pydantic	v2	Request/response data validation and models	FREE
python-socketio	5.11	WebSocket server for real-time agent progress	FREE
httpx	0.27	Async HTTP client for non-blocking API calls	FREE
Poetry	Latest	Dependency management (cleaner than pip)	FREE

2b. Background Tasks & Scheduling

Technology	Purpose	Cost
FastAPI BackgroundTasks	Run CrewAI agents async — API returns job ID immediately, no timeout	FREE
APScheduler	Trigger daily reef briefing at 7 am AEST automatically	FREE
asyncio	Async Python — built into Python 3.11, no install needed	FREE

Why this matters: CrewAI agents take 2-5 minutes. Running them synchronously would cause API timeouts. BackgroundTasks starts the crew and returns a job ID immediately. The frontend polls for status via WebSocket.

3. AI & Agent Layer

3a. LLM Configuration

Model / Tool	Used For	Speed	Cost
Llama 3.1 70B via Groq	Production — all agent reasoning, synthesis, report writing	800 tok/sec	FREE tier
Llama 3.1 8B via Ollama	Local development and testing on your machine	~30 tok/sec	FREE
Groq API	Cloud inference endpoint — 14,400 tokens/min free quota	—	FREE
Ollama	Local LLM runtime — no internet needed, no rate limits	—	FREE

3b. Agent Framework

Library	Purpose	Cost
CrewAI	Multi-agent orchestration — roles, tasks, process types	FREE
crewai-tools	Built-in agent tools (SerperDev, file tools)	FREE
LangChain	LLM interface, tool integration, retrieval chains	FREE
langchain-community	Ollama integration, document loaders	FREE
langchain-groq	Groq LLM integration for CrewAI	FREE
langchain-chroma	ChromaDB retriever integration	FREE
langchain-huggingface	HuggingFace embeddings wrapper	FREE

3c. The Six CrewAI Agents

Agent	Role	Primary Tool
Bleaching Monitor	Fetches live NOAA bleaching alerts and DHW values	NOAA ERDDAP API
Temperature Analyst	Analyses AIMS underwater temperature trends vs baselines	AIMS Logger API
News Scout	Searches for latest reef research and news	DuckDuckGo Search
Research Synthesiser	Queries RAG over indexed GBRMPA/AIMS PDF reports	ChromaDB RAG Tool
Alert Publisher	Generates zone-level public alerts with severity ratings	BOM Weather API
Report Writer	Synthesises all agent outputs into a weekly health bulletin	All above outputs

4. RAG Pipeline (All Local — No API Cost)

Component	Technology	Detail	Cost
Embeddings	sentence-transformers	all-MiniLM-L6-v2 model — 80 MB, runs on CPU, 384 dimensions	FREE
Vector DB	ChromaDB	Persistent to HF Spaces 50 GB disk — survives container restarts	FREE
PDF Parsing	pdfplumber	Extracts text from GBRMPA/AIMS PDF reports	FREE
Web Scraping	BeautifulSoup4 + lxml	Scrapes GBRMPA reef health update web pages	FREE
Chunking	LangChain TextSplitter	500-token chunks with 50-token overlap	FREE
LangChain integration	langchain-chroma	Connects ChromaDB to CrewAI agent tool	FREE

Component	Technology	Detail	Cost
HF Embeddings wrapper	langchain-huggingface	Connects sentence-transformers to LangChain	FREE

RAG Document Sources (ingested once, queried continuously):

Document Source	Type	How Ingested
GBRMPA Reef Health Updates	PDF + HTML (weekly)	BeautifulSoup scrape + pdfplumber
AIMS Annual Condition Reports	PDF (yearly)	pdfplumber
AIMS Long-Term Monitoring Data	CSV	pandas + LangChain CSV loader
Scientific papers (arXiv, AIMS)	PDF	pdfplumber
Coral bleaching event records	HTML	BeautifulSoup scrape

5. Data Sources (All Free, No Payment)

Source	Data Provided	API Key?	Python Library
NOAA Coral Reef Watch	Bleaching alerts, SST, Degree Heating Weeks — daily, 5 km res, 1985-present	None	requests + xarray
AIMS Temperature Logger	130 M underwater temp observations, 800+ reef sites, 20 years	None	requests
GBRMPA Reef Authority	Weekly reef health updates, bleaching records, zone condition data	None (scrape)	pdfplumber + BS4
Bureau of Meteorology	Weather, cyclones, El Nino/La Nina index, sea surface temp	None	requests
NASA Earthdata	Ocean colour, chlorophyll (algae blooms), SST from MODIS/VIIRS	Free account	earthaccess
Copernicus Sentinel-2	Satellite reef imagery, water quality indices	Free account	sentinelsat
ReefCloud (AIMS)	Coral cover percentages, reef imagery, open-access platform	None	requests

6. Agent Tools

Tool Name	Built With	What Agents Use It For	Cost
NOAA Bleaching Tool	requests + xarray (custom @tool)	Fetch live DHW and bleaching alert level by lat/lon	FREE
AIMS Temperature Tool	requests (custom @tool)	Fetch underwater temp time series for specific reef sites	FREE
BOM Weather Tool	requests (custom @tool)	Fetch cyclone alerts, weather anomalies near reef zones	FREE
RAG Query Tool	ChromaDB + LangChain (custom @tool)	Semantic search over all indexed reef documents	FREE
PDF Scraper Tool	pdfplumber + requests (custom @tool)	Download and parse new GBRMPA reports on demand	FREE
DuckDuckGo Search	duckduckgo-search library	Web search for latest reef news and research papers	FREE

7. Data Processing Libraries

Library	Purpose	Cost
pandas	Data manipulation, CSV handling, time series aggregation	FREE
numpy	Numerical operations, statistical calculations	FREE
xarray	Parse NetCDF4 files from NOAA (multi-dimensional arrays)	FREE
netcdf4	NetCDF4 file format support	FREE

Library	Purpose	Cost
geopandas	Spatial reef zone data and geographic operations	FREE
earthaccess	NASA Earthdata API — ocean colour, chlorophyll	FREE
sentinelsat	Copernicus Sentinel-2 satellite imagery download	FREE

8. Storage

Storage	Technology	What Is Stored	Cost
Vector Store	ChromaDB (persistent)	Reef document embeddings — survives container restarts on HF 50GB disk	FREE
Agent Memory	SQLite (built-in Python)	CrewAI long-term memory, job history, alert logs, run outputs	FREE
API Cache	JSON files on disk	NOAA/AIMS response cache — avoids redundant API calls within 1 hour	FREE
PDF Reports	Files on HF Spaces disk	Downloaded GBRMPA PDF reports before ingestion into RAG	FREE

9. Developer Tools & Code Quality

Tool	Purpose	Cost
VSCode	IDE	FREE
GitHub	Version control + portfolio visibility	FREE
Docker Desktop	Test HF Spaces Docker container locally before deploying	FREE
Postman / Thunder Client	Test and debug FastAPI endpoints	FREE
Black	Python auto-formatter	FREE
Ruff	Fast Python linter	FREE
Pre-commit	Auto-format and lint before every git commit	FREE
pytest	Test FastAPI routes and agent tools	FREE
python-dotenv	Load .env API keys into the app	FREE
loguru	Better structured logging — replaces print() statements	FREE
tenacity	Retry failed NOAA/AIMS API calls automatically	FREE

10. Environment Variables (.env)

Variable	Value	Where to Get It
GROQ_API_KEY	Your Groq API key	console.groq.com — free signup, no card
EARTHDATA_USERNAME	NASA account username	urs.earthdata.nasa.gov — free
EARTHDATA_PASSWORD	NASA account password	urs.earthdata.nasa.gov — free
COPERNICUS_USER	Copernicus account username	scihub.copernicus.eu — free
COPERNICUS_PASSWORD	Copernicus account password	scihub.copernicus.eu — free
ENVIRONMENT	production	Set manually
CORS_ORIGINS	https://your-app.vercel.app	Your Vercel URL

Variable	Value	Where to Get It
NEXT_PUBLIC_API_URL	https://your-hf-space.hf.space	Your HF Space URL
NEXT_PUBLIC_WS_URL	wss://your-hf-space.hf.space	Your HF Space URL

Note: NOAA, AIMS, GBRMPA, BOM, DuckDuckGo need no API keys at all. Total signups required: 3 (Groq, NASA, Copernicus). Zero credit cards required.

11. Project Folder Structure

Path	Description
reef-monitor/frontend/	Next.js project — deploy to Vercel
frontend/app/page.tsx	Main dashboard page
frontend/app/map/page.tsx	Full interactive reef map
frontend/app/alerts/page.tsx	Live bleaching alert feed
frontend/app/research/page.tsx	AI research Q&A interface
frontend/components/ReefMap.tsx	React Leaflet interactive map component
frontend/components/BleachingChart.tsx	Recharts temperature / DHW time series
frontend/components/AgentProgress.tsx	Live WebSocket agent progress feed
frontend/lib/api.ts	Axios API client configuration
frontend/lib/socket.ts	Socket.io client setup
reef-monitor/backend/	FastAPI project — deploy to HF Spaces Docker
backend/Dockerfile	Docker config for HuggingFace Spaces (port 7860)
backend/main.py	FastAPI entry point, routes, WebSocket server
backend/agents/crew.py	CrewAI crew assembly and kickoff
backend/agents/bleaching_monitor.py	Bleaching Monitor agent definition
backend/agents/temperature_analyst.py	Temperature Analyst agent definition
backend/agents/news_scout.py	News Scout agent definition
backend/agents/research_synthesiser.py	Research Synthesiser agent definition
backend/rag/embeddings.py	sentence-transformers embedding setup
backend/rag/vectorstore.py	ChromaDB persistent vector store
backend/rag/ingestion.py	PDF scraping + chunking + indexing pipeline
backend/rag/retriever.py	Similarity search retriever
backend/tools/noaa_tool.py	Custom CrewAI tool — NOAA bleaching data
backend/tools/aims_tool.py	Custom CrewAI tool — AIMS temperature
backend/tools/bom_tool.py	Custom CrewAI tool — BOM weather
backend/tools/rag_tool.py	Custom CrewAI tool — ChromaDB RAG query
backend/scheduler.py	APScheduler — daily 7 am AEST auto-run
backend/requirements.txt	All Python dependencies

Path	Description
backend/.env	API keys (never commit to GitHub)

12. Total Cost Summary

Component	Platform	Monthly Cost
Frontend hosting	Vercel Free Tier	$0
Backend hosting	HuggingFace Spaces (Docker)	$0
LLM inference	Groq Free Tier (Llama 3.1 70B)	$0
Local LLM (dev)	Ollama on your laptop	$0
Embeddings	sentence-transformers (local CPU)	$0
Vector database	ChromaDB (local in container)	$0
All data sources	NOAA + AIMS + GBRMPA + BOM + NASA	$0
Web search	DuckDuckGo (no API key)	$0
CI/CD	GitHub Actions (public repo)	$0
	TOTAL	$0 / month

Built for SDG 14: Life Below Water • Data: NOAA Coral Reef Watch, AIMS, GBRMPA, BOM, NASA Earthdata, Copernicus • AI: CrewAI + Llama 3.1 + RAG • Stack: Next.js + FastAPI + HuggingFace Spaces + Vercel